CALIFORNIA STATE UNIVERSITY BAKERSFIELD

Geographic Patterns of Air Pollution and Adverse Birth Outcomes in California

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Research Questions

1. Does exposure to air pollution influence birth outcomes in California counties?

2. Do counties in California's Central Valley, show different birth outcomes compared to coastal counties?

Methodology

This study used a ecological analysis of California county-level data from 2007-2023. Data collected monthly birth counts, population estimates, and air pollution concentrations (PM2.5, NO_2, Ozone, SO_2, CO) from multiple sources including the EPA and California Department of Public Health. Data was cleaned, aggregated to annual county-level averages, and merged using county names and years. Pearson correlation coefficients were calculated between air pollutants and birth outcomes, conducted regional comparisons (Central Valley vs. other areas), and performed temporal trend analysis using linear regression. Geographic visualizations were created using GeoPandas with California county shapefiles, and statistical analysis was performed using Python (3.11) with Pandas, Scipy, and Seaborn.

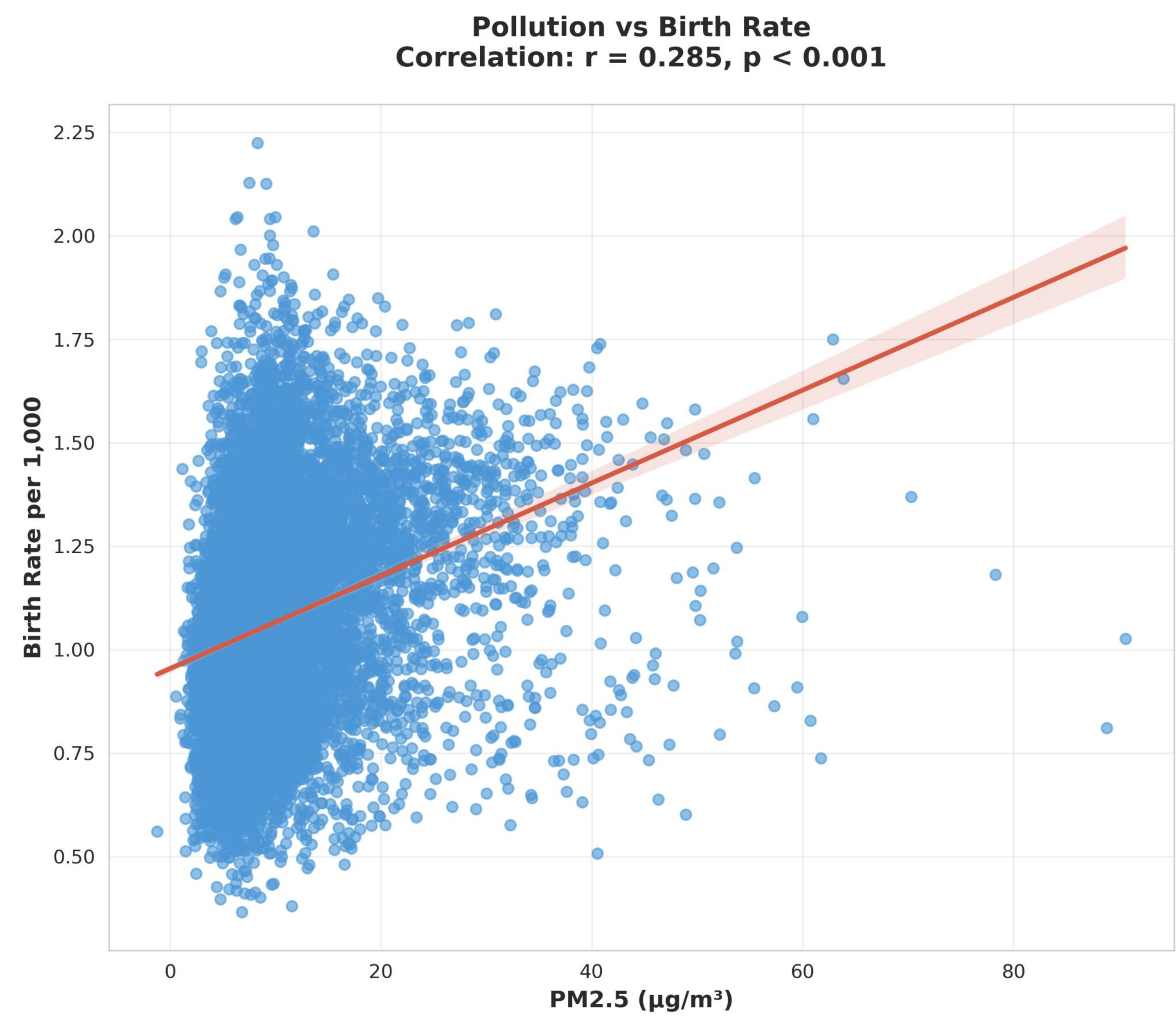
Literature Review

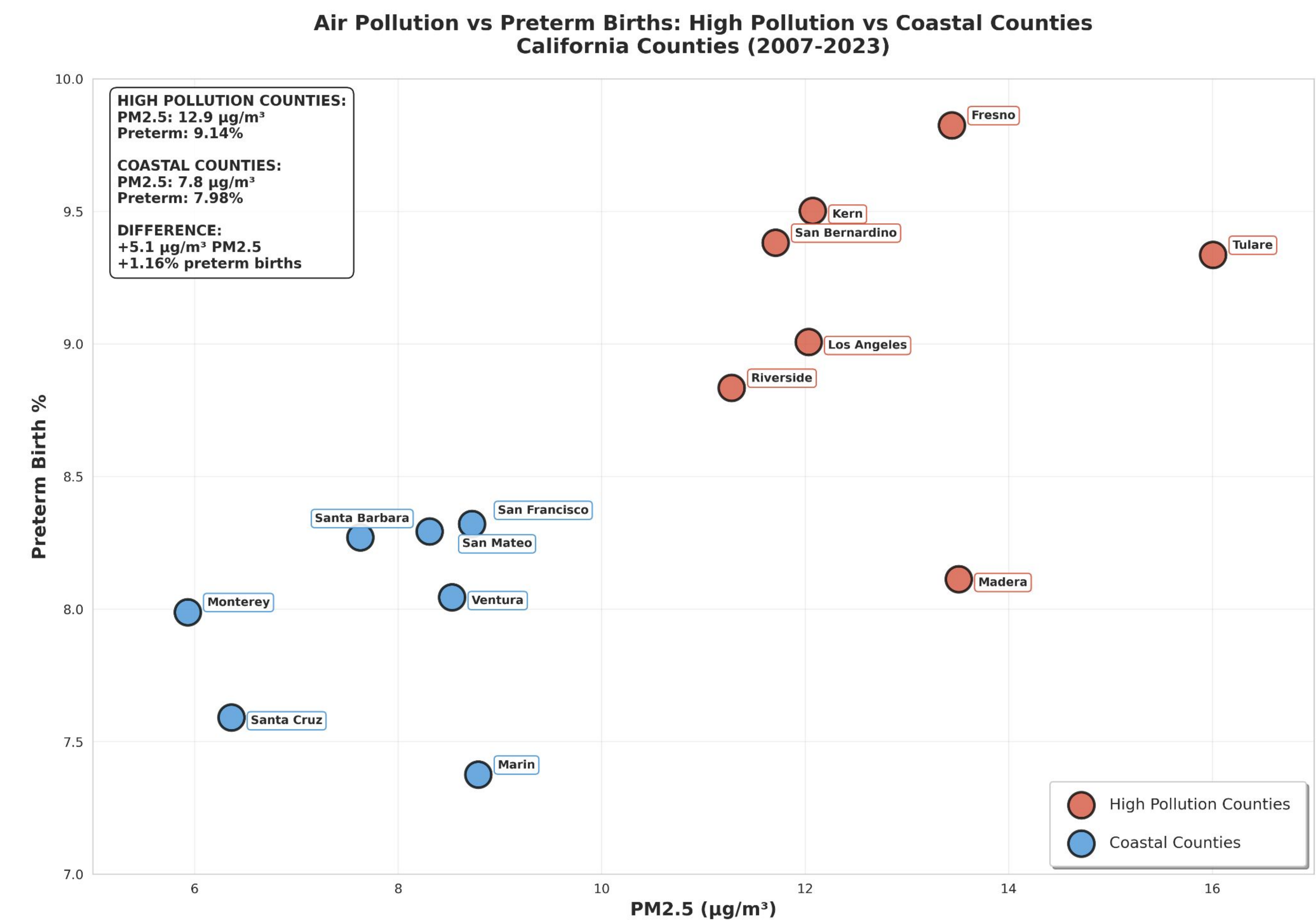
Air pollution, or specifically fine particulate matter (PM2.5) and ozone, has been widely linked to adverse birth outcomes. Multiple studies across California show that higher exposure to PM2.5 during pregnancy is associated with an increased risk of preterm birth (PTB), low birth weight (LBW), and small-for-gestational-age (SGA) infants.

Research from Los Angeles and statewide analyses also find stronger effects in lower-income and vulnerable communities. In the Central Valley, where pollution levels are typically higher, studies show that short-term increases in PM2.5 and ozone are associated with higher PTB risk.

Recent statewide research has also highlighted wildfire smoke as a growing source of PM2.5, with increased exposure linked to higher rates of PTB. These findings suggest that air pollution may trigger inflammation and oxidative stress, which can affect placental function and fetal development.

Building on this evidence, our study explores the relationship between PM2.5 levels and PTB rates across California counties, with a focus on comparing outcomes between the more polluted Central Valley and cleaner coastal regions.





Data Collection

Multiple data sources were combined to examine the relationship between air pollution and birth outcomes across California counties. Birth, population, and air quality data (2000–2023) were provided in a merged dataset. Annual county-level Air Quality Index (AQI) files for 2020–2023 were obtained from the U.S. Environmental Protection Agency (EPA). County population estimates were drawn from the U.S. Census Bureau. Preterm birth data were retrieved from the California Department of Public Health (CDPH). Geographic boundaries for California counties were obtained from U.S. Census Bureau TIGER/Line shapefiles via GeoPandas. These sources were consolidated into a primary dataset (main_dataset.csv) for county-level analyses, with preterm birth data analyzed separately.

Analysis & Results (Shown Above)

Our analysis reveals a statistically significant (~0.3 correlation) relationship between air pollution and preterm birth rates across California counties (2007-2023). While overall birth rates showed complex patterns due to demographic confounding, focusing on preterm births uncovers a clear health impact. **The 1.16 percentage point difference in preterm birth rates between high pollution and coastal counties represents a 15% relative increase in preterm births**. Given that preterm births are associated with higher infant mortality and long-term developmental issues.

High Pollution Counties (Central Valley & Inland):

- Average PM2.5: 12.9 μg/m³
- Average Preterm Rate: 9.14%
- Examples: Fresno (13.4 μg/m³, 9.8% preterm), Kern (12.1 μg/m³, 9.5% preterm)

Coastal Counties:

- Average PM2.5: 7.8 μg/m³
- Average Preterm Rate: 7.98%
- Examples: Marin (8.8 μg/m³, 7.4% preterm), Santa Cruz (6.4 μg/m³, 7.6% preterm)

References:

Bekkar et al., 2020 – systematic review across the U.S. linking PM2.5/ozone to PTB, LBW, stillbirth.
Laurent et al., 2016 – statewide California study (nested case–control, 2001–2008).
Basu et al., 2017 – California PTB and PM2.5 constituents.
Padula et al., 2014 – Central Valley (San Joaquin Valley) study on traffic pollution and PTB.
Ha et al., 2022 – San Joaquin Valley case-crossover, showing short-term ozone/PM2.5 effects on PTB.
Picciotto et al., 2024 – wildfire smoke and PTB risk in California.

https://github.com/Noah-Gallego/CV-Pathway